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TOPIC: *HOSPITAL MANAGEMENT SYSTEM*

1) Project Overview

The **Hospital Management System** is designed to automate and streamline all core operations of a hospital, including patient registration, appointment scheduling, doctor management, room and admission management, nurse assignments, and billing. The system ensures **efficient data storage**, **accurate record-keeping**, and **quick retrieval** of information to support hospital administration and patient care.

Key features include:

- Registration of patients and doctors
- Appointment booking and scheduling
- Admission and room allocation
- Nurse assignment to patients
- Bill generation and payment tracking

The database centralizes hospital information, **reduces errors**, and provides **easy access to data** for hospital staff.

2) Project Objectives

1. **Efficient Data Management:** To store all hospital-related information in a structured and secure database.
2. **Automation:** To automate manual processes such as patient registration, appointment booking, and billing.
3. **Data Accuracy:** To maintain accurate patient, doctor, nurse, and billing records.
4. **Easy Access and Reporting:** To allow quick retrieval of information for reporting and decision-making.
5. **Reduce Redundancy:** To minimize duplicate data entries and maintain data integrity.
6. **Support Hospital Operations:** To help hospital administration manage patients, rooms, staff, and financial transactions efficiently.

3) Normalization and Its Levels

Normalization is the process of organizing a database to reduce redundancy and improve data integrity. In our hospital database, normalization ensures that each table contains **only relevant data** and relationships between tables are **logical and efficient**.

Levels of Normalization Used in This Database:

1. **First Normal Form (1NF):**
    - Each table has **atomic (indivisible) values**.
    - There are **no repeating groups**.

- Example: The Patients table stores each patient as a unique row with distinct columns for name, age, gender, and contact number.

2. **Second Normal Form (2NF):**

    - **1NF is satisfied**.
    - All non-key attributes are **fully functionally dependent** on the primary key.
    - Example: In the Appointments table, date and time depend entirely on appointment_id, not partially on patient_id or doctor_id.

3. **Third Normal Form (3NF):**

    - **2NF is satisfied**.
    - There are **no transitive dependencies** (non-key attributes do not depend on other non-key attributes).
    - Example: In the Admissions table, admission_date and discharge_date depend only on admission_id and not indirectly on the room type.

How Normalization Satisfies the Database Needs:

- Eliminates **duplicate data**, e.g., patient or doctor information is stored once.
- Maintains **data integrity** through primary and foreign keys.
- Supports **scalability**, allowing new tables like Nurses or Bills without redundancy.
- Facilitates **complex queries and reporting** using joins.

4) Conclusion:

The Hospital Management System database is a **well-structured, normalized, and efficient system** that meets the operational needs of a hospital. By using 8 interrelated tables, primary and foreign keys, and normalized design (up to 3NF), the database ensures:

- Reduced data redundancy
- Accurate and consistent records
- Easy retrieval and reporting
- Support for hospital operations such as appointments, admissions, nurse assignments, and billing

Overall, this database **improves workflow, data accuracy, and decision-making** in a hospital environment, making it a **practical and essential tool** for modern healthcare management.

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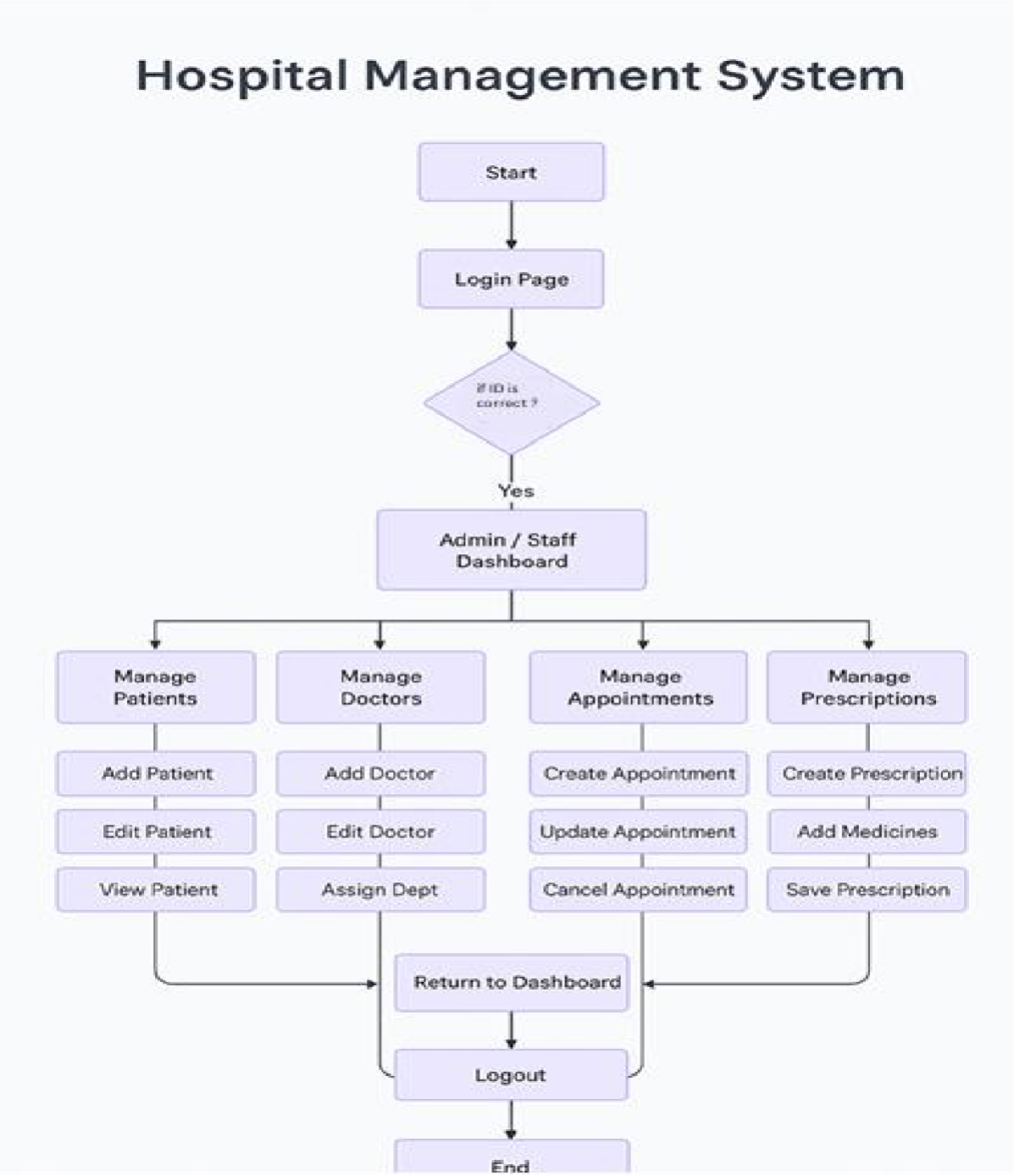
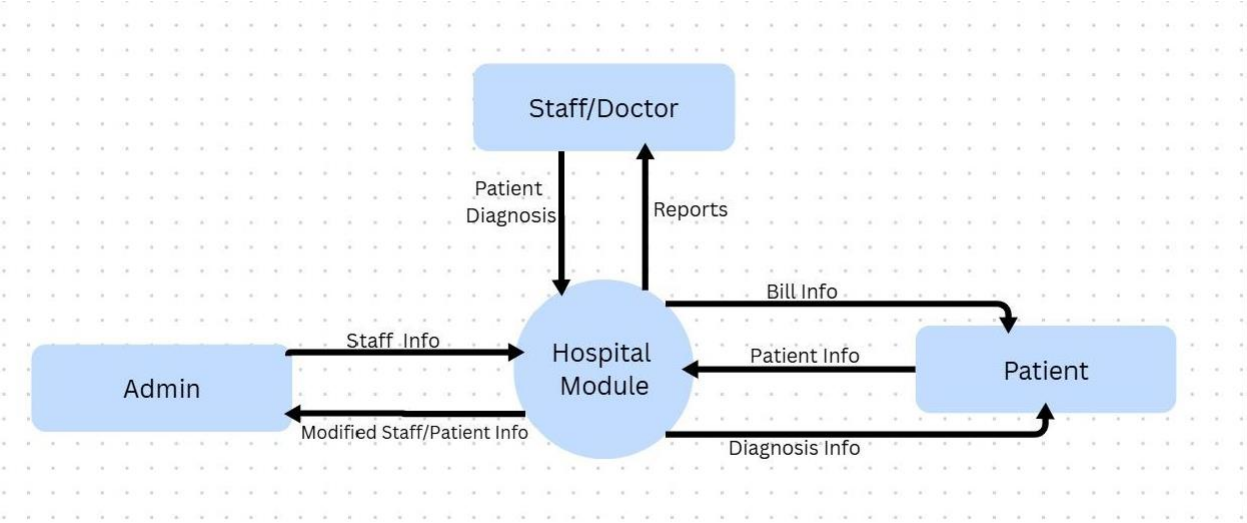
FLOW CHART

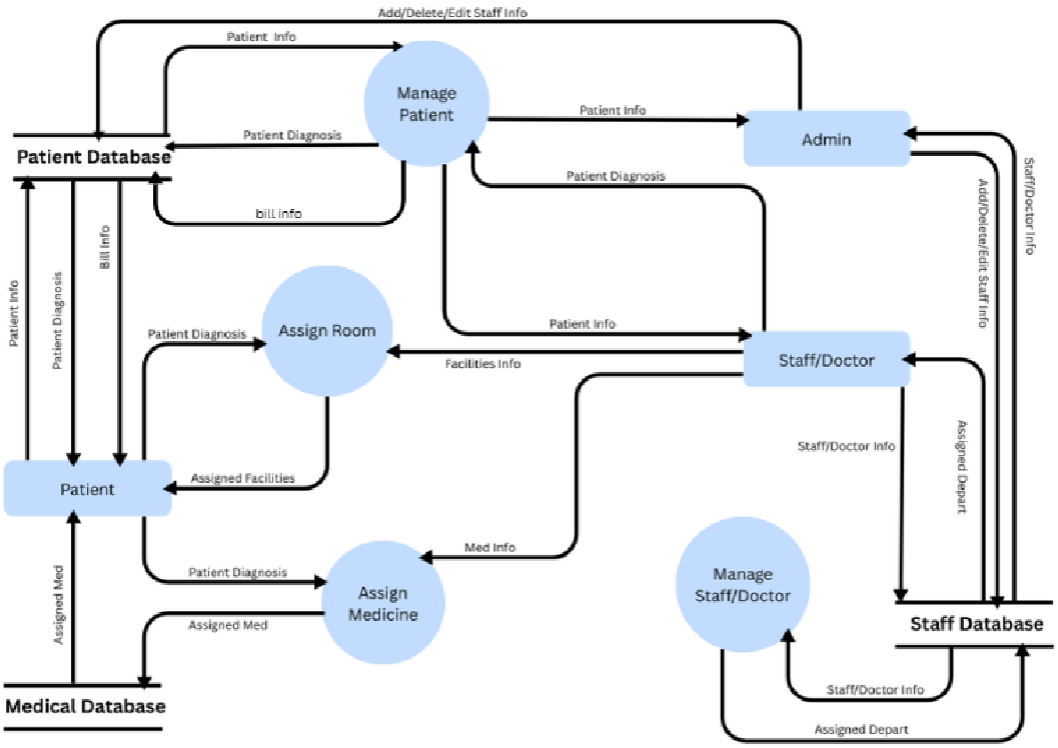

Hospital Management System
Start
Login Page
if ID is correct ?
Yes
Admin / Staff Dashboard
Manage Patients
Manage Doctors
Manage Appointments
Manage Prescriptions
Add Patient
Edit Patient
View Patient
Add Doctor
Edit Doctor
Assign Dept
Create Appointment
Update Appointment
Cancel Appointment
Create Prescription
Add Medicines
Save Prescription
Return to Dashboard
Logout
End

DATA FLOW DIAGRAM(DFD)

LEVEL 0



LEVEL 1



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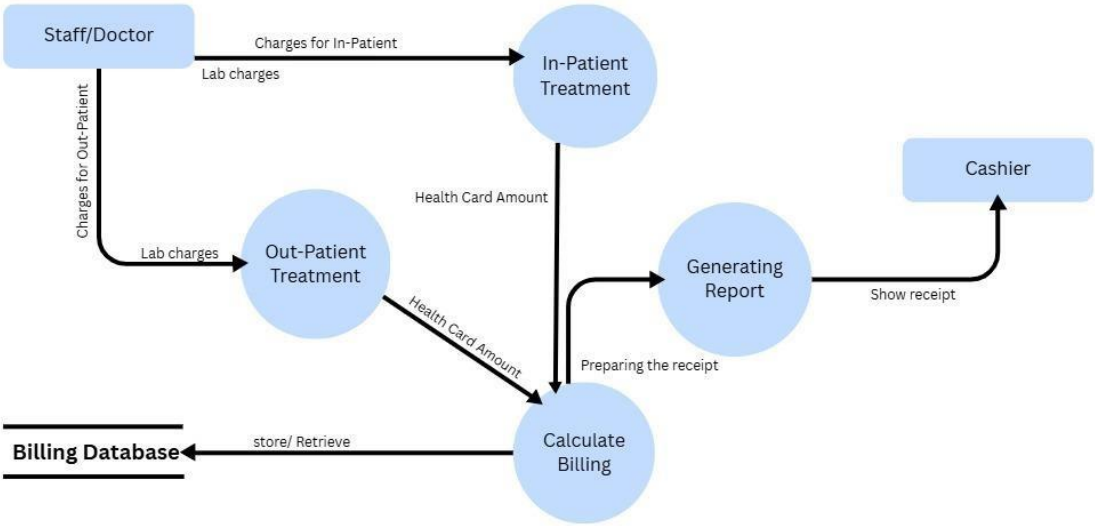
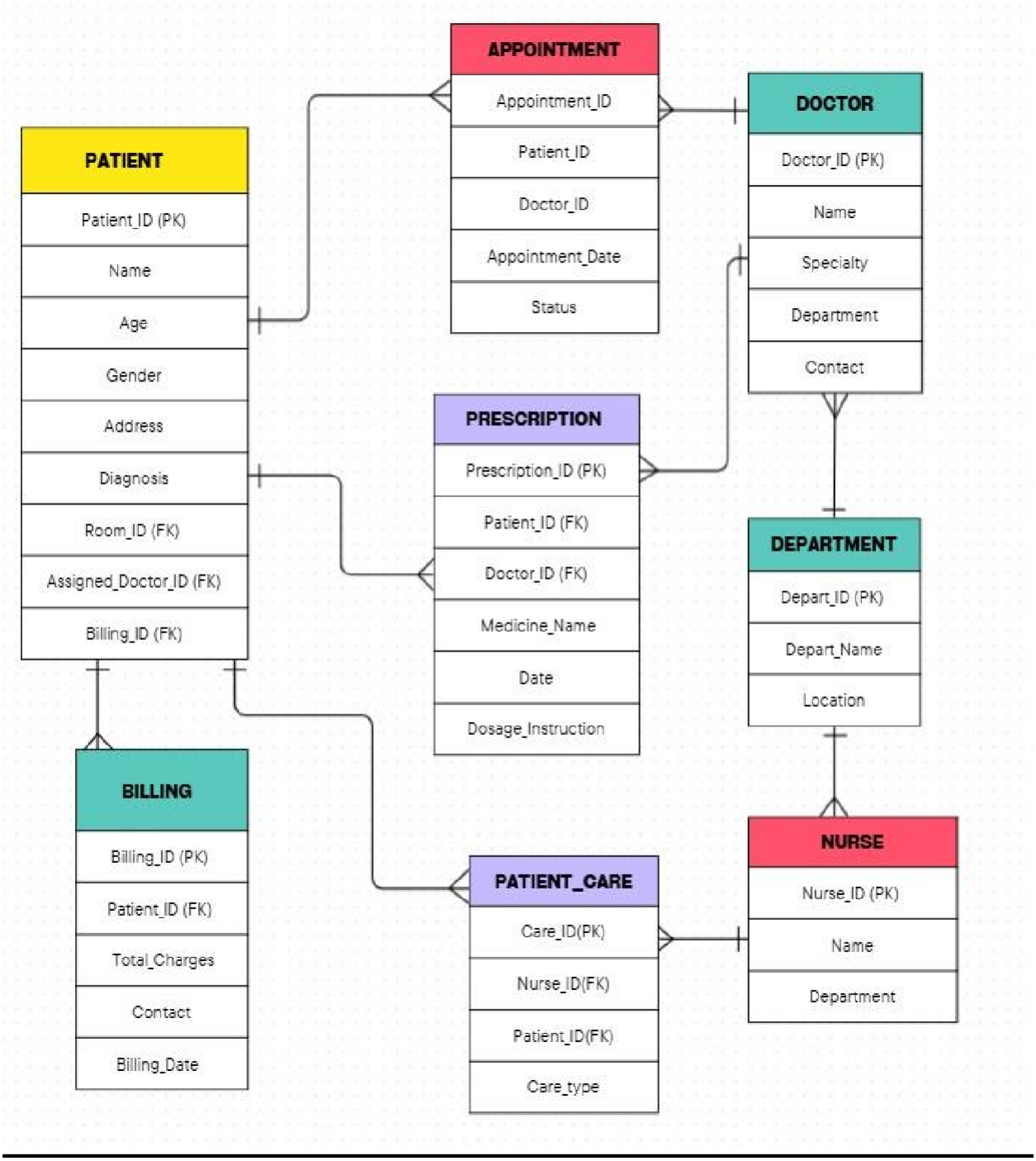
LEVEL 2


Staff/Doctor
Charges for In-Patient
Lab charges
In-Patient Treatment
Charges for Out-Patient
Lab charges
Out-Patient Treatment
Health Card Amount
Health Card Amount
Generating Report
Show receipt
Cashier
Preparing the receipt
Calculate Billing
store/ Retrieve
Billing Database

ERD:


APPOINTMENT
Appointment_ID
Patient_ID
Doctor_ID
Appointment_Date
Status
DOCTOR
Doctor_ID (PK)
Name
Specialty
Department
Contact
PATIENT
Patient_ID (PK)
Name
Age
Gender
Address
Diagnosis
Room_ID (FK)
Assigned_Doctor_ID (FK)
Billing_ID (FK)
PRESCRIPTION
Prescription_ID (PK)
Patient_ID (FK)
Doctor_ID (FK)
Medicine_Name
Date
Dosage_Instruction
DEPARTMENT
Depart_ID (PK)
Depart_Name
Location
BILLING
Billing_ID (PK)
Patient_ID (FK)
Total_Charges
Contact
Billing_Date
PATIENT_CARE
Care_ID(PK)
Nurse_ID(FK)
Patient_ID(FK)
Care_type
NURSE
Nurse_ID (PK)
Name
Department